Reg. No. | | | | | | | | | | | [14ME36]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER – 2015

THERMODYNAMICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions choosing at least two from each part.
2. Use of steam tables, Mollier diagram & thermodynamics data hand book is permitted.
3. Assume any missing data suitably.

PART-A

1. a. Define following terms: **05**

- **i.** Thermodynamic system
- **ii.** Thermodynamic state
- **iii.** Thermodynamic property
- **iv.** Thermodynamic process
- **v.** Thermodynamic equilibrium

b. Define Zeroth law of Thermodynamics? **02**

c. The readings t_A and t_B of two Celsius thermometers A and B agree at the ice point $(0^\circ C)$, and steam point $(100^\circ C)$, but elsewhere they are related by the equation $t_A = l + mt_B + nt_B^2$ where l, m and n are constants. When both thermometers are immersed in a well stirred bath, A registers $51^\circ C$ Whereas B registers $50^\circ C$. Determine the reading on B when A registers $25^\circ C$. **07**

2. a. Describe and obtain the expressions for **08**

- **i.** Spring work
- **ii.** Shaft work
- **iii.** Electrical work and
- **iv.** Magnetic work due to change in magnetization.

b. A gas is initially at 100 kPa and 600 cm^3. The final volume is 2000 cm^3, determine the moving boundary work for each of the following processes: **06**

- **i.** p is proportional to V
- **ii.** p is inversely proportional to V and
- **iii.** iii. pV^2=constant.

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3. a. State first law for a closed system undergoing a cyclic process and explain with Joule's experiment. **06**

b. A gas of mass 1.5 kg undergoes a quasi-static expansion which follows a relationship $p=a+bV$ where a and b are constants. The initial and final pressure are 1000 kPa and 200 kPa respectively and the corresponding volumes are 0.2m^3 and 1.2m^3, The specific internal energy of gas is given by the relation **08**

$u = 1.5pv - 85\ KJ/Kg$ where p is in kPa and v is in m^3/kg. Calculate the net heat transfer and the maximum internal energy of the gas attained during expansion.

4. **a.** Apply and reduce the steady flow energy equation for the following: **08**

- **i.** Steam turbine
- **ii.** Reciprocating air compressor
- **iii.** Water pump
- **iv.** Steam Nozzle.

b. A certain amount of gas is compressed from 1 bar, $0.12m^3$ to 6 bar and $0.32m^3$. The process is according to $PV^n=C$. Determine the index, magnitude and direction of work. **06**

PART - B

5. **a.** Show that two reversible engines operating between two given constant but different temperatures thermal reservoirs will have the same COP. **07**

b. A heat engine operating between two reservoirs at 1000K and 300K is used to drive a heat pump which extracts heat from reservoir at 300K at a rate twice that ate which the engine rejects heat to it. If the efficiency of the engine is 40% of the maximum possible and the COP of the heat pump is 50% of the maximum possible, what is the temperature of the reservoir to which the heat pump rejects heat? What is the rate of heat rejection from the heat pump if the rate of heat supply to the engine to 50kW? **07**

6. **a.** State and prove Clausius theorem and show that entropy is a property of the system. **08**

b. A reversible heat engine operates between two thermal reservoirs at 1000K and 300K respectively. The heat transferred to the engine is 2000KJ/Kg of heat. **06**

Verify the possibility of the development of an engine under the following heat rejections:

- **i.** Reversible
- **ii.** Irreversible
- **iii.** Imposible
  - **(i)** 600 KJ heat rejected
  - **(ii)** 900 KJ heat rejected
  - **(iii)** 300 KJ heat rejected.

7. **a.** Define the following terms: **06**

- **i.** Pure substance
- **ii.** Statured liquid
- **iii.** Saturation Pressure and
- **iv.** Critical point

b. What is quality of steam? What are the different methods of measurement of quality? **02**

c. The following observations were recorded in an experiment with a combined separating and throttling calorimeter: Pressure in the steam main – 15 bar, Mass of water drained from the separator – 0.55 kg, Mass of steam condensed after passing through the throttle valve – 4.20 kg, pressure and temperature after throttling – 1 bar, $120^{\circ}C$, Evaluate the dryness fraction of steam in the main. **06**

8. a. Define Ideal gas? Show that for a reversible adiabatic process executed by an ideal gas the following relations holds good: **07**

i. pv^{γ} = constant,

ii. $Tv^{\gamma-1}$ = constant and

iii. $Tp^{\gamma-1}$ = constant.

iv. $T\,p^{\frac{\gamma-1}{\gamma}} = C$

b. Constant volume chamber of 0.3 m^3 capacity contains 1 kg of air at $5^{\circ}C$. Heat is transferred to the air until its temperature reaches $100^{\circ}C$. Find the work done, the heat transfer, the changes in enthalpy and the changes in entropy. **07**